Advanced Visual Computing (Digital Design)

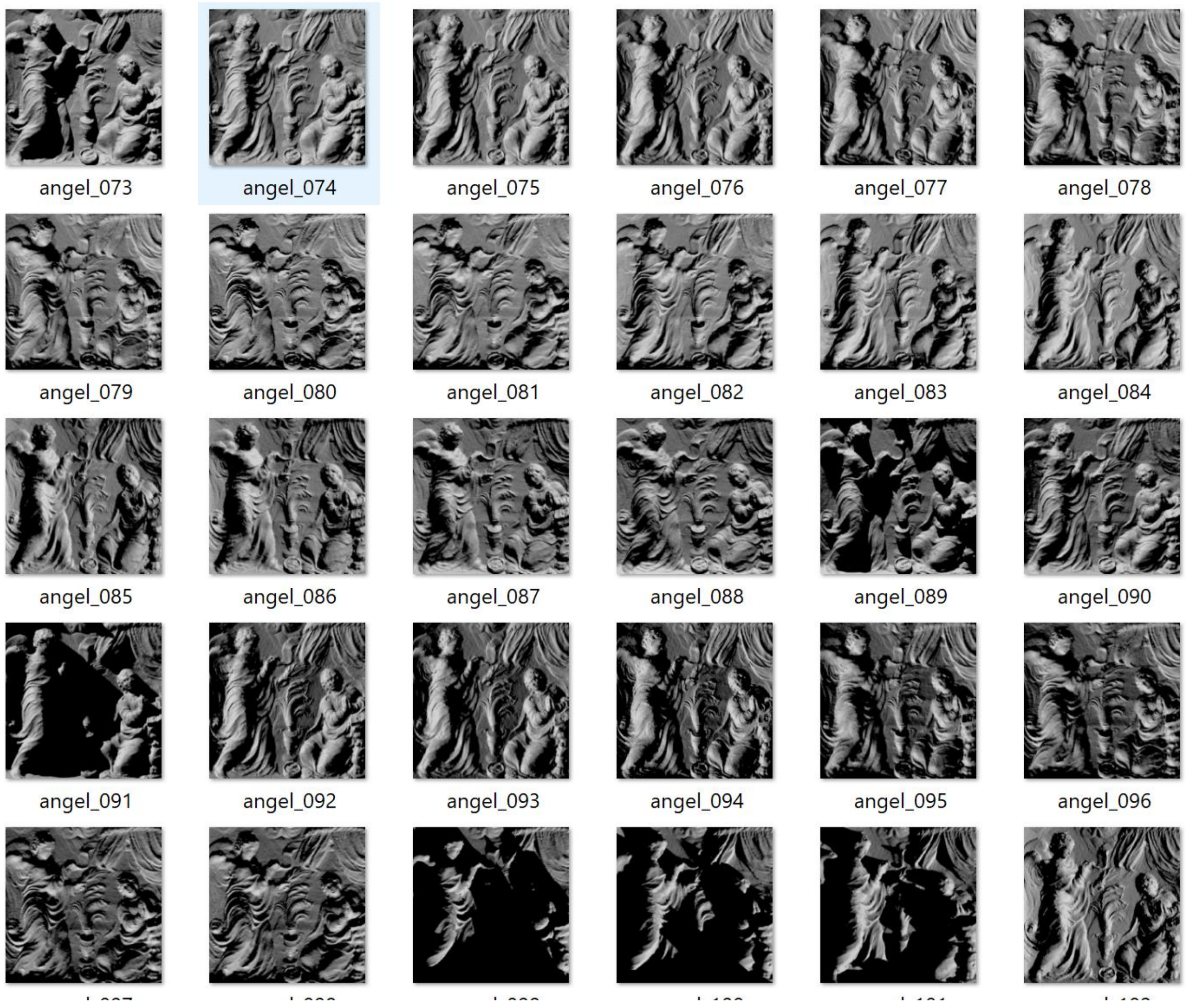
Lab 4: Photometric Stereo

Andrea Giachetti University of Verona, Italy
andrea.giachetti@univr.it

Photometric Stereo

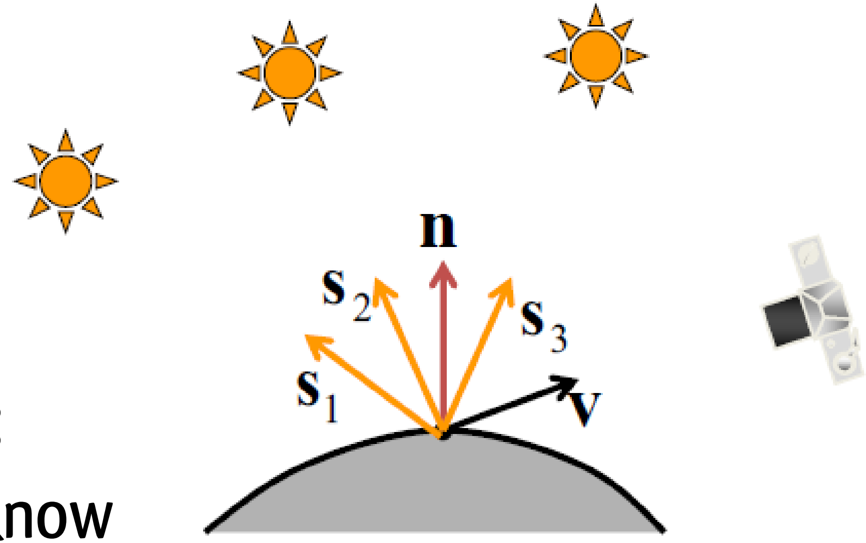
- Let us see, practically, how it works
  - Active research here on this kind of images
  - Potential topic for projects (and internship/theses)
- Lab 4 folder
  - Example synthetic and real datasets with
  - Code implementing simplest method (Lambertian)

Multi-Light Image Capture



- Calibrated: known light
- Not obvious: which light model can we use?
  - In the practice we often (not correct) use a directional or point light model and estimate the related parameters (direction/position)

Lambertian photometric stereo



Lambertian case

$$I = \frac{\rho}{\pi} kc \cos\theta_i = \rho \mathbf{n} \cdot \mathbf{s} \qquad \left(\frac{kc}{\pi} = 1\right)$$

Image irradiance:

$$I_1 = \rho \mathbf{n} \cdot \mathbf{s}_1$$
$$I_2 = \rho \mathbf{n} \cdot \mathbf{s}_2$$
$$I_3 = \rho \mathbf{n} \cdot \mathbf{s}_3$$

- We can write this in matrix form:

$$\begin{bmatrix} I_1 \\ I_2 \\ I_2 \end{bmatrix} = \rho \begin{bmatrix} \mathbf{s}_1^T \\ \mathbf{s}_2^T \\ \mathbf{s}_3^T \end{bmatrix} \mathbf{n}$$

- Simple technique but:
  - Calibration? How to know light directions?
  - If not Lambertian?
  - How to convert normals into depth maps?
    - Need to integrate

Example code

- Run the example code:
  - It shows the normal map estimated with the basic method and the corresponding ground truth for the synthetic data
    - Are them similar?
    - How can we assess this quantitavely
    - What happens if we test with different materials?
    - How could we improve the result?
  - Also includes a tentative normal integration to obtain depth
    - Not too accurate. Why?
    - Better methods are available in the literature

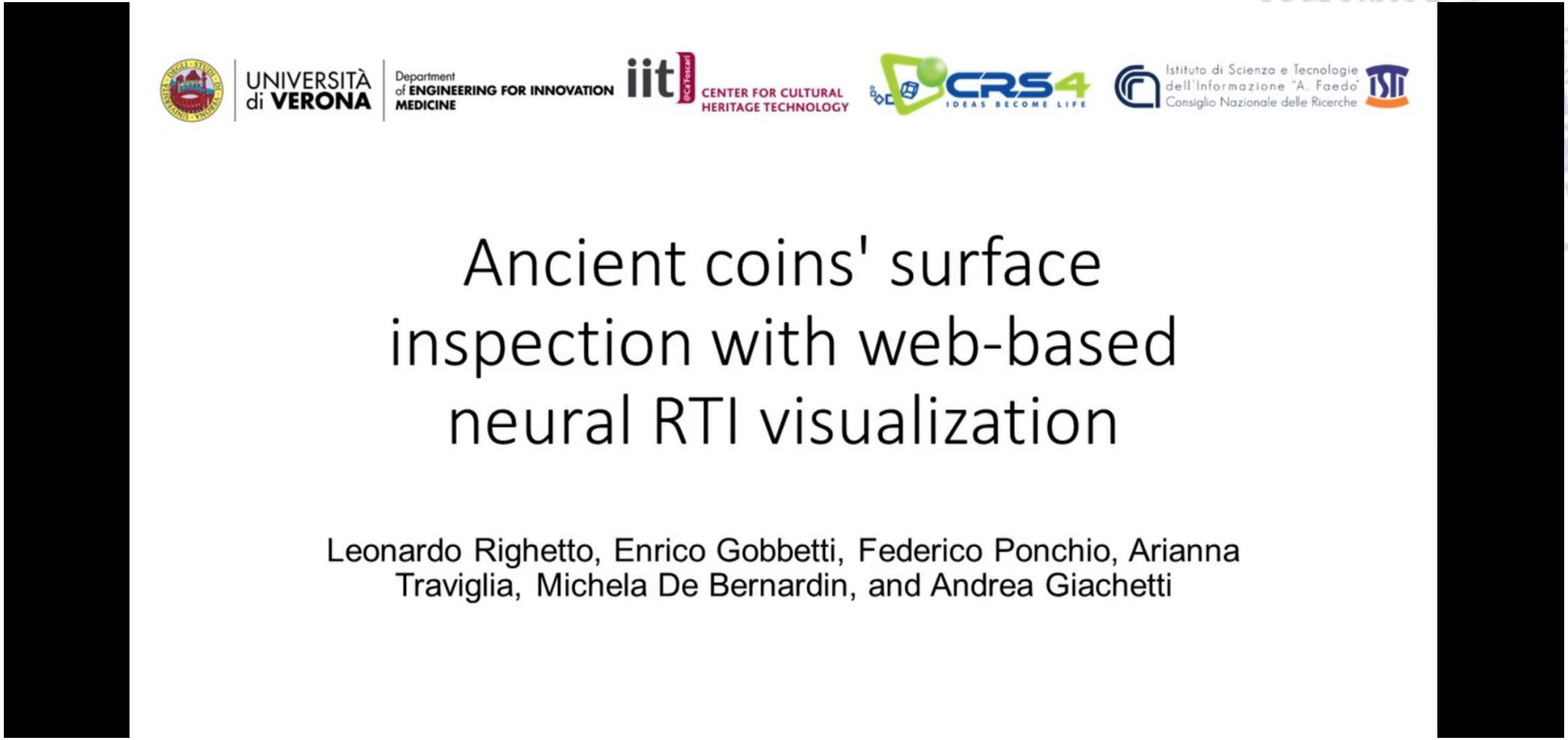
Possible improvements

- Trimming: remove highlights/shadows from the estimation
- Separate color channels
- Robust fitting:
  - Use robust kernels
  - Use Iteratively Reweighted Least Squares or other approaches

Homework: modify/evaluate

- Estimate the mean angular error between estimated and true normals
- Evaluate the average errors for all the materials and the overall mean
- Try to improve the results and submit code/report with results

Reflectance Transformation Imaging



Potential theses/internships

- Tests with novel data/polarized acquisitions
- Inverse rendering tasks: shape, depth, shadow estimation